Linked List Basics & Two-Pointer Techniques Handout

Topic: Linked List

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Resource Overview & Syllabus Scope:

Illustrated PDF guide breaking down Singly, Doubly, and Circular Linked Lists with Floyd's Cycle Detection (Hare & Tortoise), reversing in-place, and merging sorted lists.

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Linked List: Core Fundamentals & Architecture

Linked List is a critical building block within Data Structures & Algorithms.

Overview:
Illustrated PDF guide breaking down Singly, Doubly, and Circular Linked Lists with Floyd's Cycle Detection (Hare & Tortoise), reversing in-place, and merging sorted lists.

Key Concepts:
1. Foundational Architecture & Mathematical Modeling
2. Memory, Time, and Algorithmic Complexity
3. Practical Implementation Guidelines and Edge Cases.

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KEY TAKEAWAYS & CORE CONCEPTS:

- Primary Objective: Master Linked List for semester exams and technical interviews
- High-yield conceptual definitions and systematic proofs
- Optimized data structures and architectural trade-offs

```
[ Input Data / Problem Statement ]
                %
                %¼
```

EXAM FORMULAS: Time Complexity: O(log n) to O(n log n) | Space Complexity: O(1) Auxiliary or O(n) Storage

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EXAM & INTERVIEW TIP: Focus on step-by-step state tracing and edge conditions (empty inputs, single element, negative keys) during exams

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Linked List: In-Depth Analysis & Proofs

Detailed analysis of operations, runtime invariance, and algebraic proofs for Linked List.

When evaluating real-world performance, consider both worst-case and amortized guarantees. Memory layouts affect hardware cache locality and overall execution throughput.

KEY TAKEAWAYS & CORE CONCEPTS:

- Rigorous algorithmic invariant proofs
- State transitions across consecutive iterations
- Corner cases and boundary limits

```
State[i] %% (Step Transition)%% > State[i+1] %% (Convergenc
```

EXAM FORMULAS: Recurrence: T(n) = aT(n/b) + f(n) | Amortized Cost: ¦(D_i) - ¦(D_{i-1}) +

EXAM & INTERVIEW TIP: In university examinations, always state assumptions clearly before sketching state transitions.

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Linked List: Code Implementation & Solved Cases

Implementation patterns in standard systems languages (C++, Java, Python).

Key Best Practices:
- Guard against out-of-bounds index exceptions
- Validate pointer nullability and empty collections
- Avoid redundant reallocations inside nested loops.

KEY TAKEAWAYS & CORE CONCEPTS:

- Modular, maintainable code structure
- Zero memory leaks and safe pointer references
- Benchmarked execution profiles

EXAM & INTERVIEW TIP: Write comments indicating variable roles (e.g. low, mid, high) to secure full partial marking.

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Linked List: Summary & Practice Problems

Quick Revision Summary for Linked List:

Review this checklist 24 hours prior to exams:
' Definition and primary application domains
' Step-by-step dry run on at least 2 distinct sample inputs
' Complexity comparison against alternative techniques
' Typical interview trap questions and counterexamples.

KEY TAKEAWAYS & CORE CONCEPTS:

- Consolidated high-frequency viva questions
- Common student pitfalls and misconceptions
- Reference problems with verified solutions

```
[ Practice Problem ] %% > [ Identify Pattern ] %% > [ Opti
```

EXAM & INTERVIEW TIP: Re-check time allocation during exams; never spend more than 15 minutes on a 5-mark derivation.

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